order45 RESULTS: RG43 (x ↔ 1/x invariance at κ = 1)
PREREG sha256 325adfd013b59c5dd9d96bbaca293177f073951966e2c456d68521cce9496a8b (hashed before numerics).
RG43: PASS (Category B as tested; Category A candidate)
pair	c(x)	c(1/x)	rel diff	‖R(1/x)‖·x/‖R(x)‖
5/7 ↔ 7/5	0.29703878	0.29703878	3.2e−15	1.000000000000
4 ↔ 1/4	0.07617288	0.07617288	9.1e−16	1.000000000000
9/2 ↔ 2/9	0.09411413	0.09411413	4.7e−15	1.000000000000
5 ↔ 1/5	0.06496284	0.06496284	8.5e−16	1.000000000000
7/3 ↔ 3/7	0.18754287	0.18754287	9.8e−15	1.000000000000
2 ↔ 1/2	0.12614279	0.12614279	6.6e−16	1.000000000000
3 ↔ 1/3	0.09316884	0.09316884	4.3e−15	1.000000000000
2/3 ↔ 3/2	0.19197618	0.19197618	3.3e−15	1.000000000000

Max 9.8e−15 (PASS ≤ 1e−10). Prediction PASS: correct. The order44 ~1e−5 asymmetry came from κ̂ ≠ 1.
Post-hoc mechanism (after the verdict; disclosed): scripts/o45_posthoc.py
Candidate identity, with κ = 1 and Π the monomial matrix of (α,β) ↦ (α,−β) from the reduction rules:
R(1/x) = −x^{−1}·conj(Π·(35^{−1/2}M)^{−1}·R(x)), which gives ‖R(1/x)‖ = ‖R(x)‖/x.
Proof sketch:
(i) F(−x) = conj(Π·F(x)) for F = C2b. Exact: the complex conjugate of the summand e(−xQ(ν) + B(ν,β)) is e(xQ(ν) − B(ν,β)), and the B2 weights are real.
(ii) conj(kM)·Π = Π·(kM)^{−1}, with k = 35^{−1/2}.
(iii) kM is unitary.
Then R(1/x) = conj(ΠF(x)) − x^{−1}kM·conj(ΠF(−1/x)) = −x^{−1}·conj(Π(kM)^{−1}[F(−1/x) − x·kM·F(x)]).
Numerical checks: (i) 1.1e−15, (ii) 6.0e−16, (iii) 5.0e−16. The identity itself holds to 1.4e−13 (relative, at x = 2, 5/7, 9/2).
Status: (i) is proved. (ii) and (iii) are finite matrix identities verified only in floating point, not exactly. The symmetry is therefore a Category A candidate pending exact certification of (ii) and (iii) in Z[ζ_9800].
v11
No. The proof is incomplete until (ii) and (iii) are certified exactly. Ξ: not identified. E1–E3 OPEN.

order45 NOTES
- New candidate theorem: the κ = 1 residual of the C2b law satisfies R(1/x) = −x^{−1}·conj(Π(kM)^{−1}R(x)).
- Next (to preregister): exact certification of (ii) conj(M)ΠM = 35Π and (iii) M·M^† = 35I in Z[ζ_9800], using exact cyclotomic arithmetic (e.g. reduce mod Φ_9800, or an exact multiset-plus-cancellation check). If both are certified, the symmetry is Category A and warrants v11.
- The same mechanism should hold for any real-weight Bernoulli sum (C2a, C11, CO). That would be a testable prediction.
- Naming: RG* only, no labels L1–L7; "36" only as dim V. MAPS_CANDIDATES untouched; no Resonant objects; no bridge claim; CORE_FREEZE. nice, OMP_NUM_THREADS=1; no push or sends; new files only.